

L01: Why Simulate?

Simulation Without a Black Box

Outline

Learning Objectives

Three Classification Axes

Choosing the Right Tool

Jensen's Inequality and Why Randomness Matters

The 12-Step Simulation Study

In-Class Activity

Key Takeaways

Learning Objectives

By the end of this lecture you will be able to:

- Classify a system along three modeling axes: state type, uncertainty, and dynamics.
- Explain why averaging randomness produces wrong performance estimates (Jensen's inequality).
- Choose the right tool — formula, optimization, or simulation — for a given problem.
- Outline the 12-step simulation study process.

Three Axes of Model Classification

Every model lives somewhere in a three-dimensional space:

Axis	Pole A	Pole B
State space	Discrete (queue length, inventory)	Continuous (tank level, temperature)
Uncertainty	Stochastic (random arrivals)	Deterministic (fixed schedule)
Time change	Dynamic (evolves over time)	Static (snapshot, one decision)

Discrete-Event Simulation niche

DES targets systems that are **discrete**, **stochastic**, and **dynamic** — the upper-left corner that closed-form queueing theory can only approximate.

Classification Examples

Clearly DES

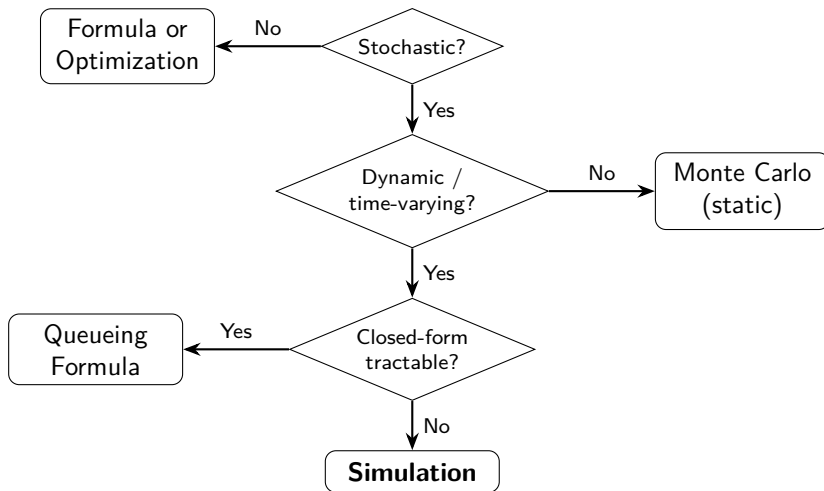
- Emergency department: patient arrivals, beds, physicians
- Port terminal: ships, cranes, trucks
- Call center: calls, agents, routing rules

Better handled otherwise

- Pipeline flow $\rightarrow$ PDE / CFD (continuous state)
- Retirement savings $\rightarrow$ Monte Carlo (static, no queue)
- Traffic macro-flow $\rightarrow$ ODE / SD (aggregate, continuous)

Choosing the wrong paradigm doesn't just add cost — it produces wrong answers.

Decision Tree: Formula vs. Optimization vs. Simulation



Use the *simplest* tool that answers the decision. Simulation is powerful but expensive to build and interpret.

Jensen's Inequality: Averaging Randomness Gives Wrong Answers

Jensen's Inequality

For a convex function f and random variable X :

$$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)]$$

Equality holds only when X is deterministic.

Queueing consequence: M/M/1 mean waiting time in queue:

$$W_q = \frac{\rho}{\mu(1 - \rho)}, \quad \rho = \frac{\lambda}{\mu}$$

Deterministic $D/D/1$ at same ρ : $W_q = 0$ (no randomness, no waiting).

Example: $\lambda = 4/\text{hr}$, $\mu = 5/\text{hr}$,
 $\rho = 0.8$

- M/M/1: $W_q = \frac{0.8}{5(0.2)} = 0.8 \text{ hr}$
- D/D/1: $W_q = 0$
- Error ratio: ∞

If you plug in average interarrival time and average service time, you get zero wait — completely wrong.

The 12-Step Simulation Study

1. Problem formulation
2. Study objectives and scope
3. Conceptual model
4. Data collection and input modeling
5. Model translation (code)
6. Verification
7. Validation
8. Experimental design
9. Production runs
10. Output analysis
11. Documentation
12. Implementation and recommendations

This course follows the same arc

Modules M01–M02: steps 1–3. M03–M05: steps 4–6. M06–M09: steps 7–12.

Activity: Classify Three Systems

For each system below, identify the three axes and the recommended tool:

1. **Urban call center** — 200 agents, time-varying call arrival rate, average handle time 4 min.
 - State: _____, Uncertainty: _____, Dynamics: _____
2. **Stock portfolio** — daily returns drawn from a distribution, hold for 1 year, compute VaR.
 - State: _____, Uncertainty: _____, Dynamics: _____
3. **Campus heating loop** — continuous fluid temperature, fixed pump schedule, known heat loss.
 - State: _____, Uncertainty: _____, Dynamics: _____

Discuss: which one is closest to the DES sweet spot? Why?

Key Takeaways

- Simulation is not a universal tool — use the three-axis classification to justify it.
- Randomness compounds non-linearly: replacing random quantities with their means systematically underestimates congestion and delay (Jensen's inequality).
- The right workflow has 12 steps; the code is step 5 of 12.
- DES is the paradigm for discrete, stochastic, dynamic systems — the environment where queueing formulas break down.

Next lecture: System taxonomy and choosing the right modeling paradigm (L02).